Model Complexity Versus Transportability in Ovarian Cancer Risk Prediction: A Cross-Cohort Benchmark with Decision Analysis

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Abstract

We test whether model complexity translates into transportable clinical performance for preoperative ovarian-cancer risk stratification—the serum-marker triage of the most lethal gynecologic malignancy—under external distributional and structural shift. Eight models—a one-parameter CA125 rule, two clinical formulas (ROMA, CPH-I), an unbinned logistic regression, an integer scorecard with seven whole-point weights, and three tree ensembles—were trained on a Chinese cohort ($n = 349$) and applied frozen to two external Asian cohorts (West China, $n = 380$; Japan, $n = 177$). On the primary external validation cohort (West China), the scorecard (AUROC 0.929) and the unbinned regression (0.936) were statistically indistinguishable, and each exceeded the remaining models (ROMA, CPH-I, and the tree ensembles); pooled across both cohorts, the scorecard beat ROMA (+0.043, $p = 0.006$) and all ensembles (+0.046 to +0.054, $p \leq 0.002$). The published cutpoints missed 82 of 188 cancers versus 47 for the scorecard's high-risk tier. A simulation calibrated to the training data provides a plausible mechanistic account in which *concept drift*—structural change in the marker-to-outcome mapping—produces the observed ranking. For small-sample clinical ML, external validation and drift-aware simulation distinguish what transfers from what merely fits.

Keywords: ovarian cancer; risk prediction; integer scorecard; external validation; transportability; distribution shift; decision curve analysis

Data and Code Availability The three cohorts are public, de-identified datasets archived at Mendeley Data (Chinese training cohort, https://doi.org/10.17632/th7fztbrv9.11), figshare (West China, https://doi.org/10.6084/m9.figshare.28831256), and Karger figshare (Japan, https://doi.org/10.6084/m9.figshare.24235450). All analysis code, figure scripts, and benchmark outputs are available in an anonymized code repository included as supplemental material with this submission; the repository will be de-anonymized and made public at https://github.com/angelayaoo/Ovarian-Cancer-RiskSLIM for the camera-ready version. The repository includes a one-command pipeline (`reproduce.sh`) that regenerates every table and figure from the public cohorts, with pinned dependencies and all intermediate results archived.

Institutional Review Board (IRB) This is a secondary analysis of fully de-identified, publicly archived data; no IRB approval is required in the originating jurisdictions, and the originating studies obtained their own approvals in line with the Declaration of Helsinki.

1. Introduction

Preoperative risk assessment decides which women with an ovarian mass are referred to oncologic surgery, and it leans on two serum markers, CA125 and HE4, combined into fixed formulas such as ROMA (Moore et al., 2011) and the Copenhagen Index CPH-I (Karlsen et al., 2015). Machine-learning models have been proposed repeatedly for this task (Kawakami et al., 2019; Lu et al., 2020), usually with strong internal performance and little external evidence. The broader clinical-prediction literature has established three uncomfortable facts: flexible models seldom beat well-regularized logistic regression on tabular data, especially at the sample sizes of single hospitals (Christodoulou et al., 2019; van der Ploeg et al., 2014); prediction models frequently fail to transfer across settings (Roberts et al., 2021; Wynants et al., 2020); and interpretable models belong in high-stakes medicine unless clearly worse (Rudin, 2019; Caruana et al., 2015). What is missing is a systematic account of *when* complexity repays its cost, quantified across the dimensions that matter for deployment: external discrimination, calibration, decision consequences, robustness to measurement error, sample-size efficiency, and structural drift.

Prior work shows flexible models often fail to beat logistic regression (Christodoulou et al., 2019) and that prediction models degrade across settings (Roberts et al., 2021; Wynants et al., 2020), but rarely quantifies *when* complexity pays: across what sample sizes, distribution shifts, and structural changes. We address this gap with three public cohorts and a pre-specified protocol. Contributions: (i) an eight-model benchmark with selection and preprocessing frozen on the training cohort, so external performance measures transportability; (ii) a decision-analytic evaluation—operating points, weighted-harm analysis, decision curves—showing where published cutpoints fail; (iii) a synthetic simulation reproducing the empirical ranking under a structural-shift account; and (iv) a factor decomposition quantifying each shift component's contribution to the model gap. The protocol itself is the transferable artifact for small-sample clinical ML.

2. Related Work

Reviews find no benefit of ML over logistic regression on tabular data (Christodoulou et al., 2019), and simulation studies show modern techniques are "data hungry" (van der Ploeg et al., 2014; Riley et al., 2020). RiskSLIM learns sparse integer scoring systems with certified optimality (Ustun and Rudin, 2016); bedside-computable scores persist for this reason (Caruana et al., 2015; Rudin, 2019). We combine paired bootstrap and DeLong tests, meta-analytic pooling, calibration, and decision curves (DeLong et al., 1988; Hanley and McNeil, 1983; DerSimonian and Laird, 1986; Van Calster et al., 2019; Vickers and Elkin, 2006) into one protocol, adding a drift-aware simulation.

3. Setup

Data. Three public, de-identified Asian cohorts, archived at Mendeley Data (Chinese; https://doi.org/10.17632/th7fztbrv9.11), figshare (West China; https://doi.org/10.6084/m9.figshare.28831256), and Karger figshare (Japan; https://doi.org/10.6084/m9.figshare.24235450). The target population is women with a suspected ovarian mass and preoperative CA125 and HE4 measured. Chinese training cohort: $n = 349$ (171 ovarian cancers, 178 benign masses), 49 variables. West China: $n = 380$ (188 cancers, 192 benign cysts); HE4 missing in 74%. Japan: $n = 177$ (27 cancers, 150 healthy women); age missing in 85%. Because Japan contrasts cancers with healthy controls rather than benign masses, it serves as an *intentional* case-mix stress test rather than a second triage validation: transportability claims should be probed at the limits of their case mix, and a screening-like cohort is that limit. Features: CA125 (U/mL), HE4 (pmol/L), age, menopausal status. External missing values are replaced with per-feature training medians, frozen; no statistic is computed on a validation cohort. The primary endpoint is the scorecard-versus-ROMA AUROC difference on West China; all other analyses are secondary. The imputation strategy is varied in the no-HE4 and complete-case sensitivity analyses (Results). Per-feature Kolmogorov–Smirnov shifts versus the training distribution range 0.13–0.38 (West China, mean 0.22) and 0.43–0.60 (Japan, mean 0.50).

Models. (1) *Integer scorecard*: continuous features rank-transformed, cut into three equal-frequency bins, fitted with L2-regularized logistic regression, coefficients rescaled and rounded to integers in $[-5, 5]$: each coefficient is rescaled as $w_i = \mathrm{round}(\beta_i \times 5/\max_j |\beta_j|)$, with $|\beta_i| < 0.01$ set to zero. The configuration is selected strictly internally by the one-standard-error rule over a pre-specified 12-configuration grid (bins $\times$ regularization strength), with all preprocessing refitted inside each training fold. (2) *Unbinned logistic regression* on standardized features. (3–4) *ROMA* (Moore et al., 2011) and *CPH-I* (Karlsen et al., 2015), published formulas computed on the same median-imputed raw features as every other model. (5) *CA125 $\geq$ 35 U/mL rule*. (6–8) *XGBoost* (Chen and Guestrin, 2016) (100 trees, maximum depth 6, learning rate 0.3), *CatBoost* (Prokhorenkova et al., 2018) (1,000 iterations, depth 6, learning rate 0.03), and *Random Forest* (Breiman, 2001) (100 trees, no depth cap) at package defaults; a 27-configuration tuning grid and nested cross-validation are reported as sensitivity analyses. Every model uses the same four features and the same frozen preprocessing; models are trained once on the Chinese cohort and applied unchanged everywhere else.

Evaluation protocol. *Discrimination*: AUROC per cohort; pairwise comparisons by 2,000 patient-level paired bootstrap resamples (two-sided empirical p-values); random-effects pooling across the two external cohorts (DerSimonian–Laird (DerSimonian and Laird, 1986)) with I^2 as the heterogeneity measure. *Internal validity*: five-fold stratified CV (StratifiedKFold, shuffle, seed 42) repeated 20 times; in-sample AUROC to quantify memorization; nested CV (outer five-fold, inner five-fold over a 9-configuration grid per family) for the tuned ensembles. *Calibration*: Brier score; calibration-in-the-large, computed as $\log(\bar{p}/(1-\bar{p})) - \log(\bar{y}/(1-\bar{y}))$; and LOWESS-smoothed calibration curves (fraction 0.8, clipped to $[0,1]$). *Decision analysis*: sensitivity/specificity at published cutpoints; weighted-harm analysis in which, for exchange-rate weights $w \in \{1,2,5,10,20,50\}$ (harm of one missed cancer relative to one unnecessary referral), the threshold minimizing $H = w \cdot \text{missed} + \text{unnecessary}$ per 100 women is identified for each model; decision curves with bootstrap bands (Vickers and Elkin, 2006). *Robustness*: Monte Carlo multiplicative noise on CA125/HE4 (0–30%, 100 perturbations per level), additive and combined noise, 5% gross-error contamination, ±10% cutpoint perturbation, and learning curves retraining on $n = 60$–349. *Simulation*: a generative model calibrated to the training data. Class-conditional biomarkers are sampled as lognormal mixtures, $z_k \sim \mathcal{N}(m_k^{(y)}, s_k^{(y)})$, with class means and variances estimated per class from the Chinese cohort's log-transformed CA125 and HE4 and raw age; the outcome follows a Bernoulli model with frozen-standardized features,

$$\begin{aligned}\eta(\mathbf{z}) &= -0.65 + 1.8\,z_1 + 1.0\,z_2 + 0.5\,z_3 \\ &\quad + (1-d)\big(1.6\,z_1 z_2 + 1.2\sin(1.7\,z_1)\big),\end{aligned} \tag{1}$$

where z_1, z_2, z_3 are the standardized biomarkers and $d \in \{0,1\}$ is concept drift (under drift the interaction and nonlinear terms vanish from the test-side outcome model). Factors: training size $n \in \{100, 200, 400, 800\}$; distribution shift implemented as shrinkage of class-conditional means toward the pooled mean by $\delta \in \{0, 0.5, 1.0\}$ (plus variance inflation), so discrimination decreases monotonically in δ; multiplicative measurement noise on CA125/HE4, $\sigma \in \{0, 0.5\}$ (additive Gaussian on the log scale), applied only to the test set; 20 replicates per cell; test $n = 4{,}000$. Models are trained on clean, stationary data and evaluated frozen on the shifted test set, mirroring the real-cohort protocol. Implementation: Python 3.13, scikit-learn 1.6, XGBoost 2.1, CatBoost 1.2, fixed seeds (random_state = 42).

4. Results

We report external discrimination first, then internal validity and calibration, decision consequences, robustness, and the simulation.

4.1. External discrimination

Discrimination on the primary external validation cohort. On West China, the scorecard (0.929, 95% CI 0.902–0.955) and the unbinned logistic regression (0.936, 0.908–0.961) were statistically indistinguishable ($p = 0.48$). Both exceeded CPH-I (0.868), ROMA (0.886), and the tree ensembles (0.879–0.887) (West column, Table 1); in the Japan column CPH-I was highest (0.920). Figure 1 shows the per-model values.

Table 1: AUROC by model and cohort at 0% measurement noise. Train = five-fold cross-validated (CV) AUROC on the Chinese training cohort; West and Japan are bootstrap means on the external cohorts. Bold marks the cohort-wise best fitted model. [a] Published fixed formula (applied value, not fitted); ROMA = Risk of Ovarian Malignancy Algorithm; CPH-I = Copenhagen Index.

Model	Train (CV)	West	Japan
Scorecard	0.906	**0.929**	0.843
Logistic regr.	0.894	**0.936**	0.804
ROMA[a]	0.912	0.886	0.797
CPH-I[a]	0.912	0.868	**0.920**
CA125 rule[a]	0.727	0.642	0.781
XGBoost	0.882	0.887	0.734
CatBoost	0.900	0.879	0.749
Random Forest	0.895	0.879	0.792

Pooled external comparison. Across both external cohorts (Table 2), the scorecard exceeded ROMA by +0.043 (95% CI 0.013–0.074, $p = 0.006$, $I^2 = 0$). It exceeded every tree ensemble by +0.046 to +0.054 (all $p \leq 0.002$) and matched the unbinned regression (-0.005, $p = 0.55$). I^2 was 0 for ROMA and for each tree ensemble.

Heterogeneity. The CPH-I comparison was the only heterogeneous one ($I^2 = 79\%$): the differ-

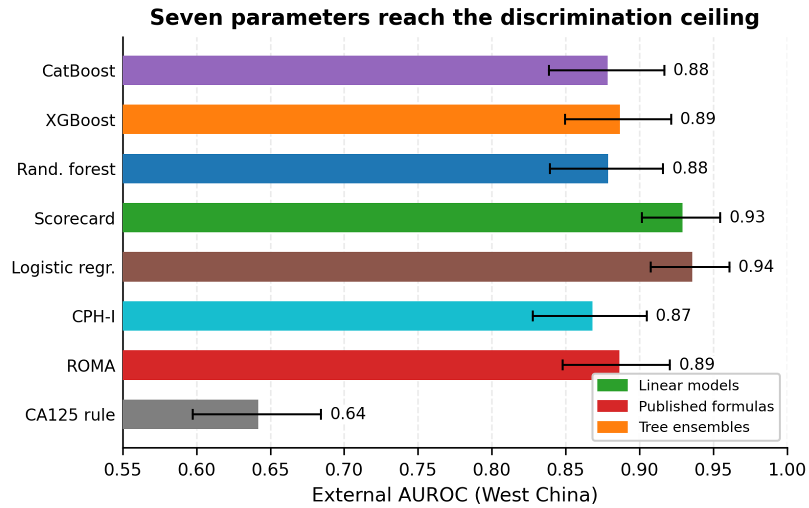

Figure 1: External AUROC on the West China cohort (n = 380; 188 cancers, 192 benign), with models ordered left to right by increasing complexity. Points are bootstrap means (2,000 resamples) with 95% confidence intervals; values are printed above the points. Markers encode model family: circles = linear models, squares = published formulas, triangles = tree ensembles. ROMA = Risk of Ovarian Malignancy Algorithm; CPH-I = Copenhagen Index.

Table 2: Scorecard-minus-comparator AUROC differences. Δ West and Δ Japan are paired bootstrap differences (2,000 patient-level resamples) on the West China cohort (n = 380; 188 cancers, 192 benign) and the Japanese stress-test cohort (n = 177; 27 cancers, 150 healthy women). Pooled = DerSimonian–Laird random-effects estimate across the two cohorts; CI = 95% confidence interval; I^2 = heterogeneity statistic; p = two-sided empirical p-value from the paired bootstrap. AUROC = area under the receiver operating characteristic curve.

Comparator	Δ West	Δ Japan	Pooled	95% CI	p	I^2
Logistic regression	−0.007	+0.039	−0.005	−0.023 to 0.012	0.554	0%
ROMA	+0.043	+0.046	+0.043	0.013 to 0.074	0.006	0%
CPH-I	+0.061	−0.076	+0.005	−0.127 to 0.137	0.942	79%
CA125 rule	+0.288	+0.064	+0.186	−0.032 to 0.404	0.094	89%
XGBoost	+0.043	+0.109	+0.046	0.017 to 0.074	0.002	0%
CatBoost	+0.051	+0.095	+0.054	0.024 to 0.084	$<$ 0.001	0%
Random Forest	+0.051	+0.051	+0.051	0.022 to 0.080	$<$ 0.001	0%

ence was +0.061 in West China and −0.076 in the Japanese cohort (27 cancers, 150 healthy women). In that cohort, 86% of healthy women and 30% of cancer patients fell into the lowest training-calibrated bins of both markers.

Internal validity. Across 20 repeated five-fold CVs the scorecard's mean internal AUROC was 0.906 (SD 0.003); the deployed three-bin configuration was selected in 18 of 20 repeats. In-sample, the ensembles fitted at 0.991–1.000 and fell by 0.094–0.117 under CV (scorecard: 0.005); nested CV of tuned ensembles reached 0.885–0.903.

4.2. Internal validity and calibration

Calibration. On West China (prevalence 49.5%), Brier scores were 0.097 (recalibrated scorecard), 0.111 (unbinned regression), 0.128 (XGBoost), 0.209 (ROMA), and 0.259 (CPH-I). Calibration-in-the-large was −0.95 (ROMA) and −1.42 (CPH-I) (Rolfsen et al., 2020).

4.3. Decision consequences

Decision consequences. At its high-risk tier (score $\geq +1$) the scorecard achieved 75.0% sensitivity and 97.4% specificity, missing 47 of 188 cancers with 5 false-positive referrals. At their published cutpoints ROMA achieved 56.4% sensitivity and 96.9% specificity (82 cancers missed, 6 false positives); CPH-I achieved 74.5% and 94.8% (48 missed, 10 false positives). In the weighted-harm analysis (Table 3, Figure 2), the unbinned regression had the lowest minimum harm at every exchange rate. The scorecard was within 2.4–7.6 harm points per 100 women of it and below ROMA at all rates $\leq$ 10:1. A second analysis applied the same within-cohort recalibration to the published formulas: minimum harms were 12.1 (scorecard), 14.7 (ROMA), and 15.0 (CPH-I) at $w = 1$, and 45.0, 45.8, and 49.5 at $w = 10$; at $w = 50$, recalibrated ROMA (45.8) was below the scorecard (59.5). Decision curves with bootstrap bands placed the recalibrated scorecard highest across thresholds 0.10–0.60.

4.4. Robustness and data efficiency

Measurement noise. Under 30% multiplicative noise on CA125/HE4 the scorecard lost 0.040 AUROC (−4.3%), versus −0.002 for the unbinned regression, −0.074 (ROMA), and −0.086 (XGBoost) (Figure 3). The ordering was the same under additive noise, combined noise, and 5% gross-error contamination. Shifting the frozen tertile cutpoints by ±10% changed the scorecard's AUROC by at most 0.016.

Table 3: Minimum net harm per 100 women, $H = w \cdot \text{missed} + \text{unnecessary}$, at each exchange-rate weight w (West China). Lower is better; bold marks the best model per column.

Model	w=1	w=5	w=10	w=50
Logistic regression	**9.7**	**29.2**	**39.7**	**46.6**
Scorecard (recal.)	12.1	30.5	45.0	59.5
ROMA	15.0	43.4	45.8	45.8
CPH-I	15.0	51.3	64.7	159.5
XGBoost	14.7	44.7	66.1	215.5
Treat all	50.5	50.5	50.5	50.5

Data efficiency. In the learning curves (Figure 4), the unbinned regression reached 0.925 at $n = 60$ and 0.936 at $n = 349$; the scorecard 0.901→0.929; the ensembles never exceeded 0.90 at any training size.

Missing-data sensitivity. For patients whose HE4 or age was median-imputed, evaluation used the remaining complete features. On the 100 patients with measured HE4 (63 cancers), every model's AUROC was at least as high as on the full cohort (Table 4). Refitting the models without HE4 changed the scorecard's West China AUROC by 0.004 (0.925), the unbinned regression's by 0.004 (0.932), and XGBoost's by +0.004 (0.891).

Table 4: Complete-case sensitivity on West China: AUROC restricted to the 100 patients with measured HE4 (63 cancers), versus the full cohort ($n = 380$; median-imputed HE4).

Model	Full (n=380)	Complete (n=100)
Scorecard	0.929	0.971
Logistic reg.	0.936	0.978
ROMA	0.886	0.965
XGBoost	0.887	0.923

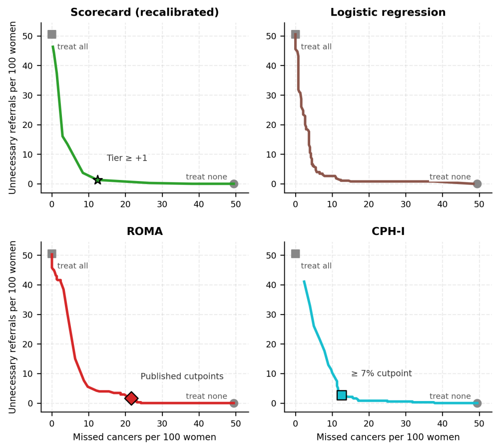

Figure 2: Decision-consequence trade-off curves (West China), one panel per model: missed cancers vs unnecessary referrals per 100 women as the decision threshold is swept over 1–100%. Grey markers are the treat-none (circle) and treat-all (square) references. The scorecard panel marks its high-risk tier (≥+1); the ROMA and CPH-I panels mark their published cutpoints.

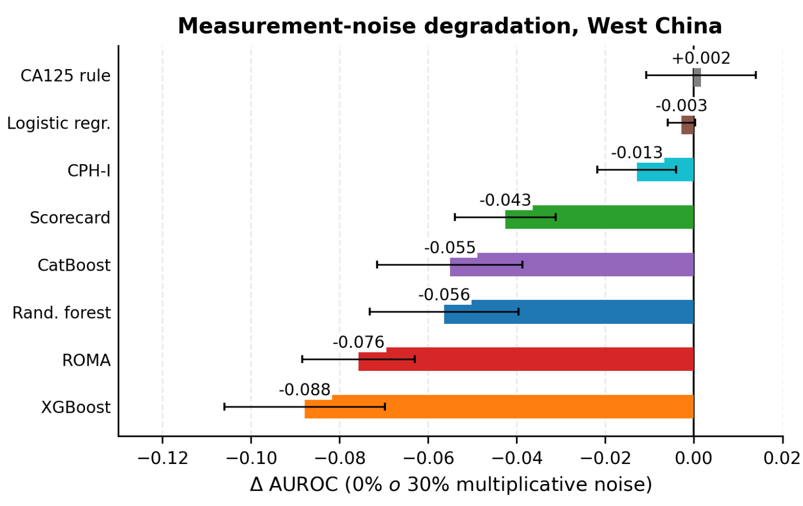

Figure 3: Change in external AUROC on the West China cohort ($n = 380$) from 0% to 30% multiplicative noise on CA125/HE4. Bars are mean changes over 100 independent perturbations; whiskers are ±1 SD across perturbations.

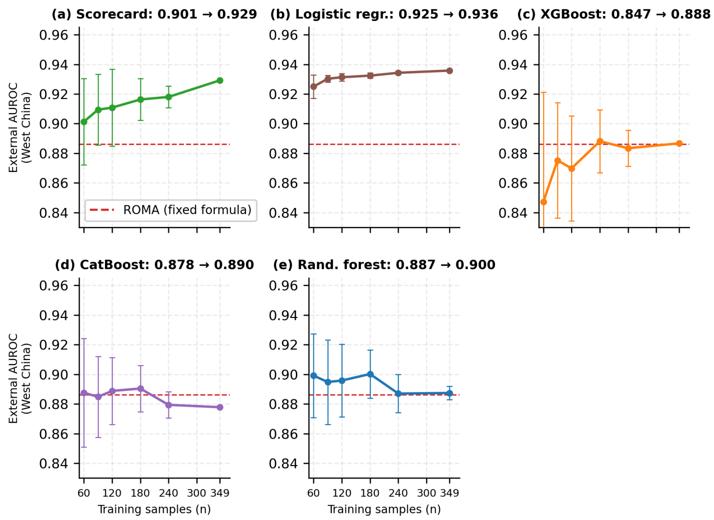

Figure 4: External learning curves on West China ($n = 380$), one panel per model: models refitted on stratified training subsamples of size n
in
$60, 90, 120, 180, 240, 349$ and evaluated frozen on West China. Points are means ± SD over 10 draws; the dashed line marks ROMA's fixed-formula AUROC. Panel titles give each model's minimum-to-maximum external AUROC over the training sizes.

4.5. Simulation

Simulation. Under stationary structure, CatBoost exceeded the scorecard by 0.01–0.05 across cells, and the purely linear regression trailed by 0.06–0.21. Under concept drift, the scorecard ranked first in 22 of 24 cells; the ensembles were 0.01–0.04 below it, and the unbinned regression was within 0.02–0.07 of it (Figure 5). In the factor decomposition (Table 5), the gap spread across drift levels was 0.033. The spreads were 0.003 for training size, 0.012 for distribution shift, and 0.002 for noise. At partial drift ($d = 0.5$) the XGBoost-minus-scorecard gap was -0.016 (between $+0.009$ at $d = 0$ and -0.025 at $d = 1$); the CatBoost gap was $+0.024$, $+0.006$, and -0.006 at $d = 0, 0.5, 1$. The empirical cohorts had mean KS shifts of 0.22–0.50 relative to training.

Table 5: Factor decomposition of the XGBoost-minus-scorecard test AUROC gap in the simulation (factor levels: drift 0/1, shift 0–1, size 100–800, noise 0/0.5). The spread across drift levels (0.033) was larger than across any other factor (0.002–0.012). Under stationary structure CatBoost won all 24 cells; under drift the scorecard won 22 of 24.

Factor	Gap range	Spread
Concept drift	$+0.007 \to -0.026$	**0.033**
Distribution shift	$-0.015 \to -0.003$	0.012
Training size	$-0.011 \to -0.008$	0.003
Measurement noise	$-0.010 \to -0.009$	0.002

4.6. Reclassification and sensitivity analyses

Reclassification. The integrated discrimination improvement of the scorecard's frozen map was $+0.071$ (95% CI 0.036–0.106) versus ROMA, $+0.110$ versus CPH-I, -0.048 versus the unbinned regression, and -0.197 versus XGBoost; the category-free NRI

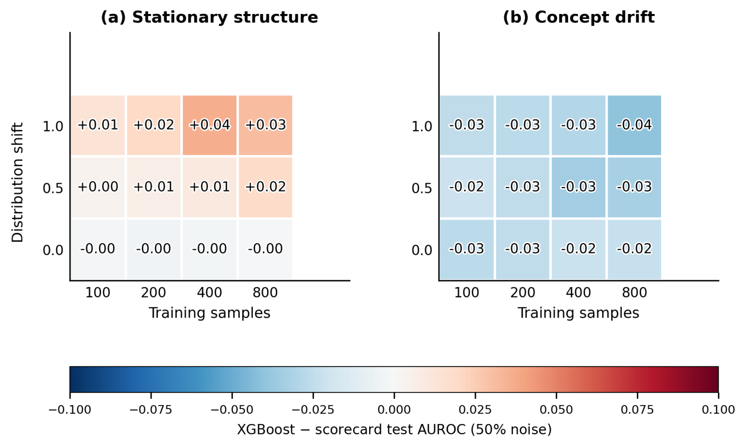

Figure 5: Regime map from the synthetic study: cell values are mean XGBoost-minus-scorecard test AUROC across 20 replicates per cell (50% measurement noise), by training size and distribution shift. (a) Stationary structure. (b) Concept drift.

versus ROMA was −0.23 (event +0.68, non-event −0.91) (Pencina et al., 2008).

Sensitivity analyses. Across the 12 L2 selection candidates, five-fold CV ranged 0.900–0.913. The configuration maximizing internal CV (4 bins, CV 0.913) achieved 0.880 externally; the selected three-bin configuration (CV 0.907) achieved 0.929. L1 variants reached 0.907 externally and two-bin configurations 0.847. The tuned tree ensembles (best of 27 configurations per family) reached 0.890 (XGBoost), 0.877 (CatBoost), and 0.894 (Random Forest) on West China. A training-fitted KNN imputation of the Japanese missing values (outcome-correlated) gave 0.907; the frozen-median convention was retained.

5. Discussion and Limitations

Three findings stand out. First, at $n < 400$ the discrimination ceiling is reachable with seven integer parameters; further degrees of freedom *cost* external performance, extending (Christodoulou et al., 2019). Second, the published cutpoints missed 82 of 188 cancers at a prevalence (49.5%) far from the Western development prevalence—base-rate dependence (Rolfsen et al., 2020)—and a locally calibrated scorecard dominates them at all rates $\leq$ 10:1. Third, the simulation offers a plausible mechanistic account in which *concept drift* produces the observed ranking; the factor decomposition and nonlinearity-strength sweep show it is graded rather than an artifact, and it is a falsifiable probe rather than a demonstration that drift occurred. For deployment, the chosen model class determines how a tool behaves when population and assay platform change—a failure mode internal evaluation cannot reveal. The protocol—frozen external cohorts, meta-analytic pooling, decision consequences, a drift-aware simulation—is proposed as a default template for small-sample clinical ML (Collins et al., 2015).

Scope. The benchmark is serum-only because the public cohorts lack imaging variables; ultrasound-based models (RMI, IOTA ADNEX) are the standard of care where imaging is available, and no superiority over them is claimed. Findings apply to serum-based triage, where ultrasound is unavailable or a first-stage screen.

Limitations: the external cohorts exhibit observed distribution shift (per-feature KS statistics 0.22–0.50 relative to training), whereas concept drift is a property of the simulation—it is not directly measured in the real data, and the simulation's role is to show that drift *can* produce the observed ranking, not that it did. The Japanese stress-test cohort has 27 cancers (∼20% power for a 0.05 difference); because its controls are healthy women, its specificity estimates are not interpretable as triage specificities; it is used only to bound discrimination under case-mix shift. Single geographic region; prospective validation pending.

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Appendix A. Supplementary Details

Deployed scorecard. Integer weights (CA125/HE4/age/menopause): bins $[-1, 0, +2]$ / $[-4, -1, +5]$ / $[-2, 0, +2]$ / $[0, 0, 0]$; score range -7 to $+9$. Tertile cutpoints in training units: CA125 24.3/165.0 U/mL, HE4 45.8/92.7 pmol/L, age 37.0/52.0 years.

Nonlinearity-strength sweep. We test whether the regime map depends on the hand-crafted outcome model by repeating the sweep at four strengths $\gamma \in \{0.4, 0.8, 1.6, 2.4\}$ of the injected interaction/sinusoidal terms of Equation (1) ($n = 200$, 50% noise, 20 replicates). The mechanism is graded rather than knife-edge: the ensembles' stationary-regime advantage grows monotonically with γ (CatBoost $+0.01$ to $+0.05$), and under drift it erodes monotonically, crossing to a scorecard advantage of -0.01 to -0.04 once $\gamma \geq 1.6$; at $\gamma \leq 0.8$ no class has an appreciable edge in either regime. The conclusion requires only learnable nonlinear structure that drifts, not a specific generative model. Ensemble-minus-scorecard gaps (mean over shift cells):

Table 6: Ensemble-minus-scorecard gaps across nonlinearity strengths.

	CatBoost − scorecard		XGBoost − scorecard	
γ	Stationary	Drift	Stationary	Drift
0.4	+0.007	+0.005	−0.013	−0.017
0.8	+0.008	+0.007	−0.008	−0.013
1.6	+0.028	−0.006	+0.014	−0.026
2.4	+0.050	−0.017	+0.038	−0.035